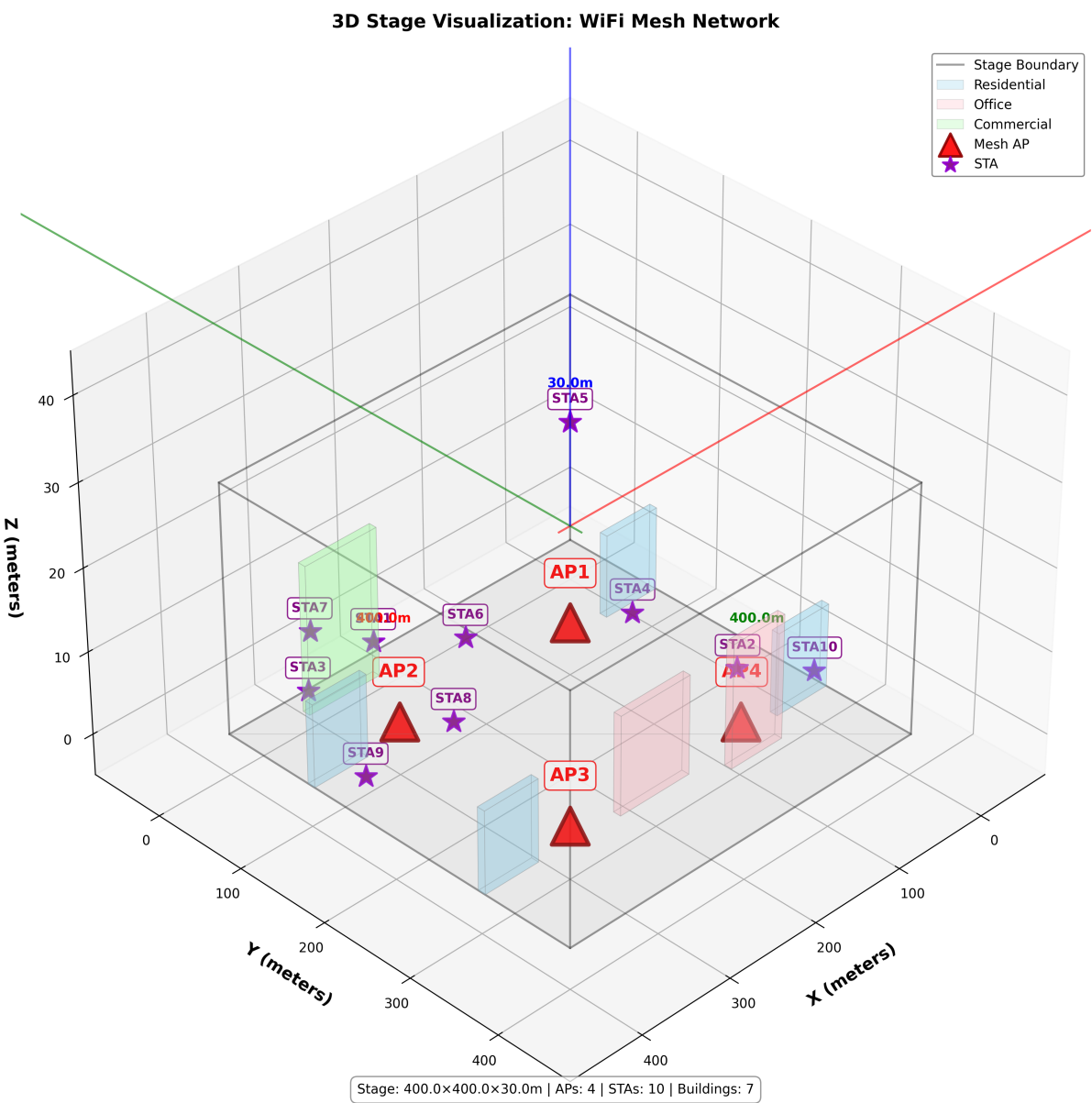


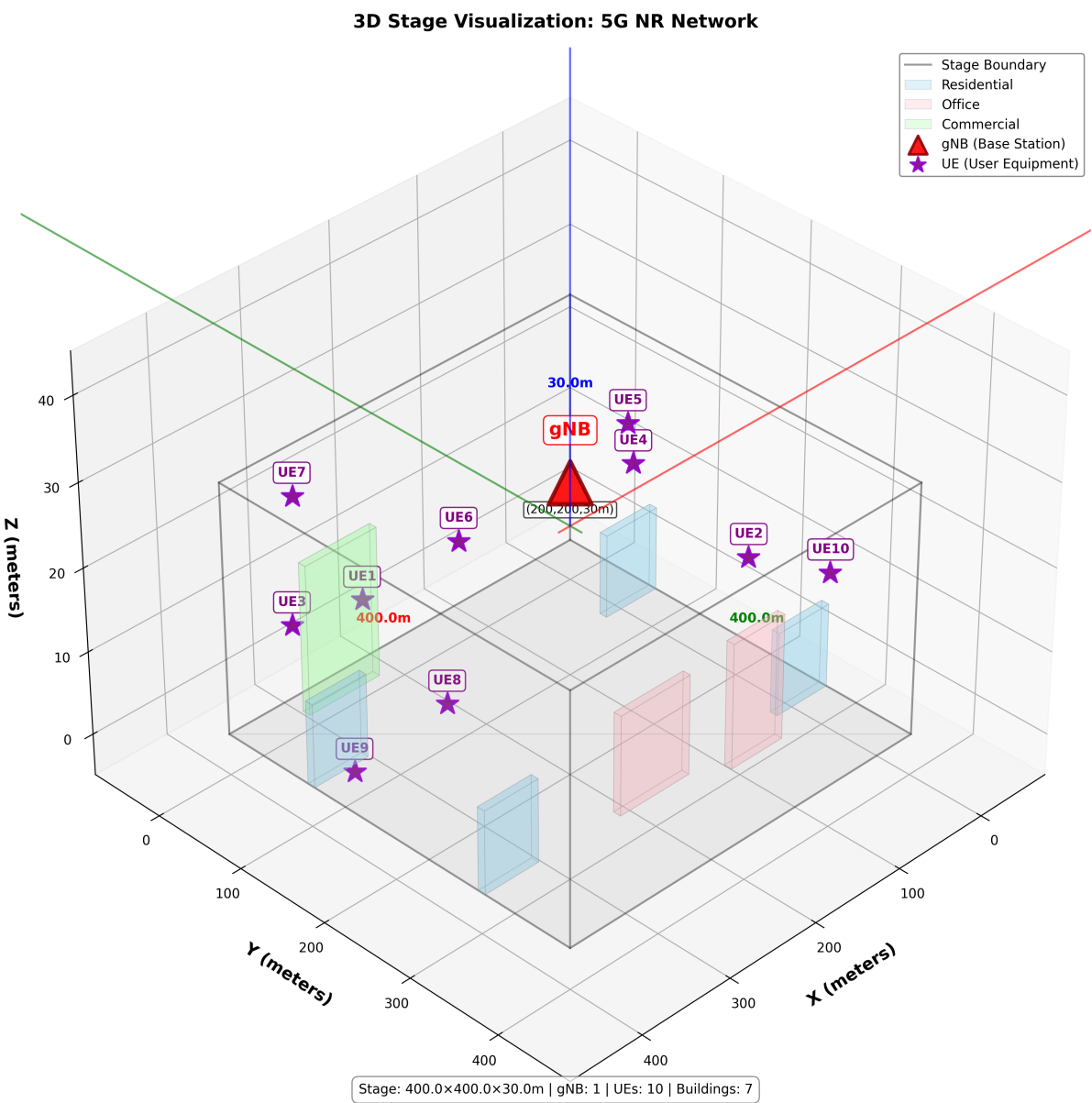
Cross-Technology Network Performance Analysis: 4G, 5G, and WiFi Mesh

- Scenarios analyzed: 108
- Generated on: 2025-11-14 22:05:20

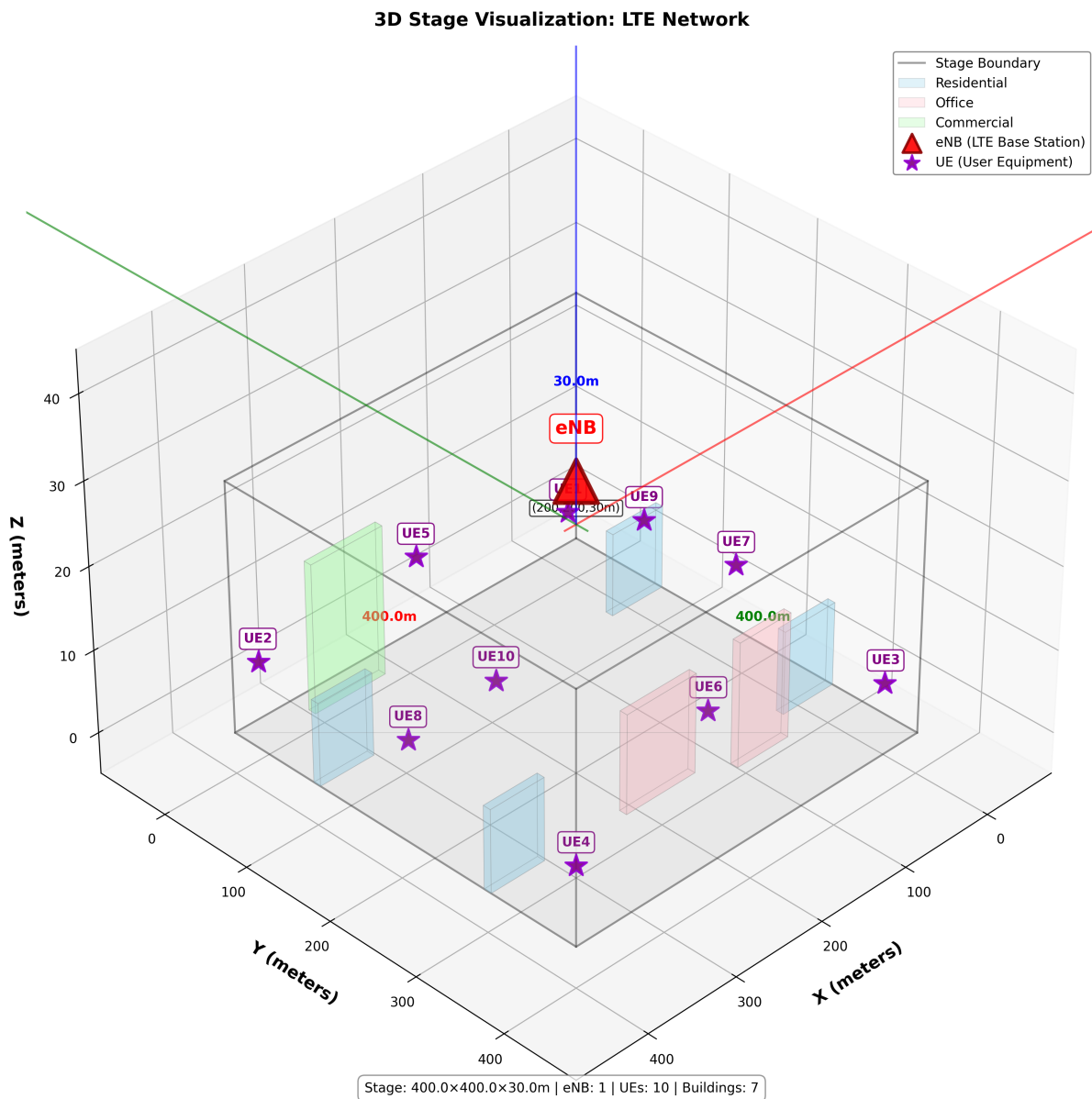
WiFi Mesh Network Visualization



5G Network Visualization



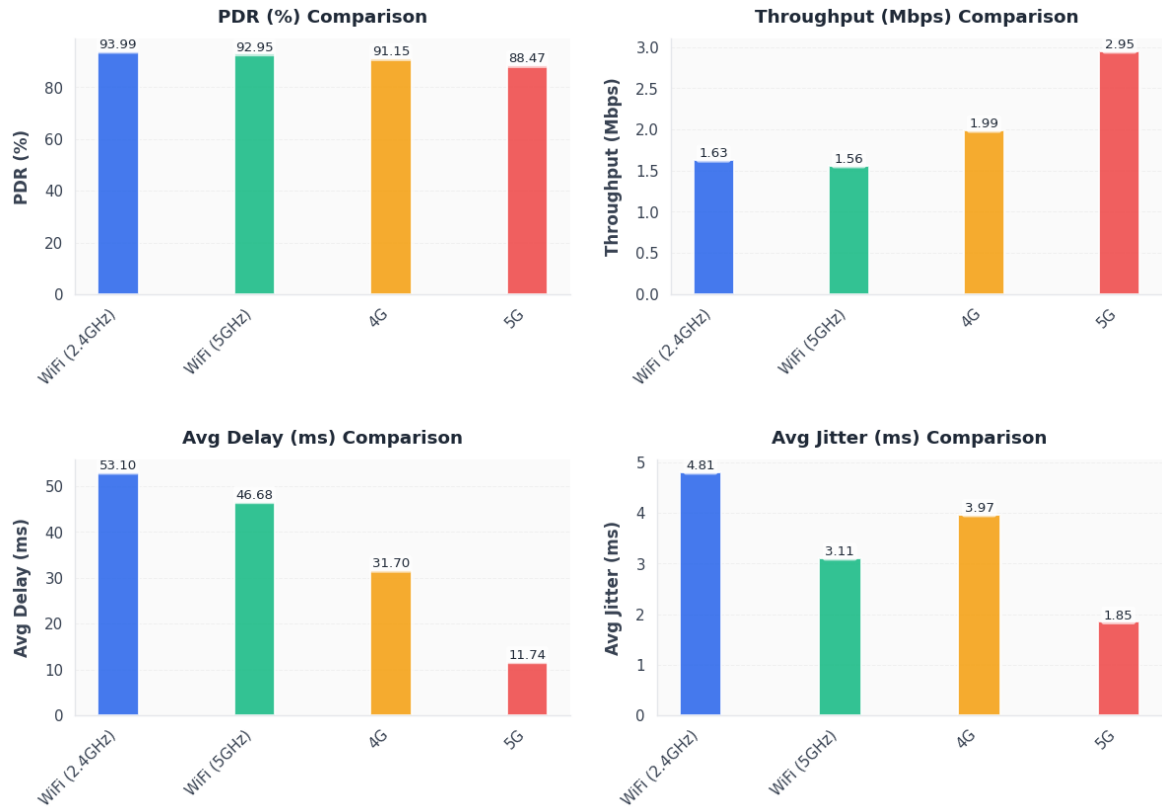
4G Network Visualization



1. Cross-Technology Performance

- 5G achieves the highest Throughput (Mbps) (2.95).
- WIFI Mesh (2.4 GHz) achieves the highest PDR (%) (93.99).
- 5G achieves the lowest Avg Delay (ms) (11.74).

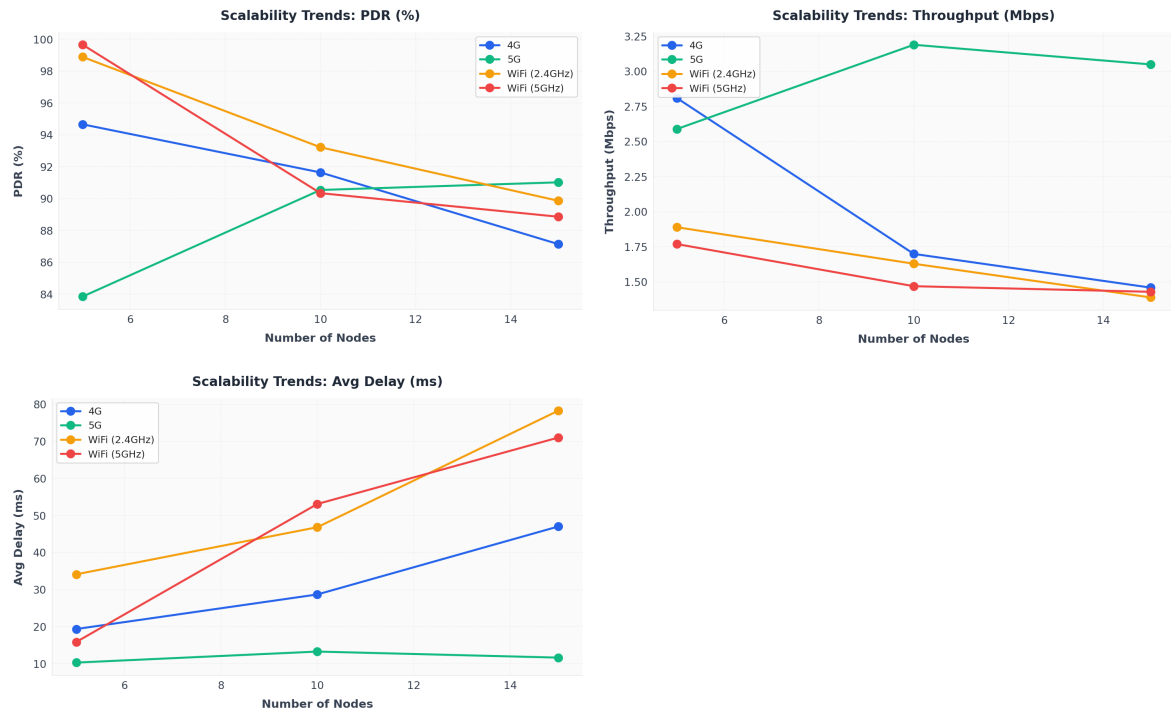
Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)	Mean Jitter (ms)
WIFI Mesh (2.4 GHz)	93.99	1.63	53.10	4.81
WIFI Mesh (5 GHz)	92.95	1.56	46.68	3.11
4G	91.15	1.99	31.70	3.97
5G	88.47	2.95	11.74	1.85



2. Scalability Trends

- WIFI Mesh (2.4 GHz) PDR changes by 9.04 points and delay by 44.18 ms between 5 and 15 nodes.
- WIFI Mesh (5 GHz) PDR changes by 10.79 points and delay by 55.19 ms between 5 and 15 nodes.
- 4G PDR changes by 7.52 points and delay by 27.67 ms between 5 and 15 nodes.

- 5G PDR changes by 7.17 points and delay by 1.34 ms between 5 and 15 nodes.



5 Nodes

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	98.89	1.89	34.13
WIFI Mesh (5 GHz)	99.66	1.77	15.86
4G	94.66	2.81	19.36
5G	83.85	2.59	10.30

10 Nodes

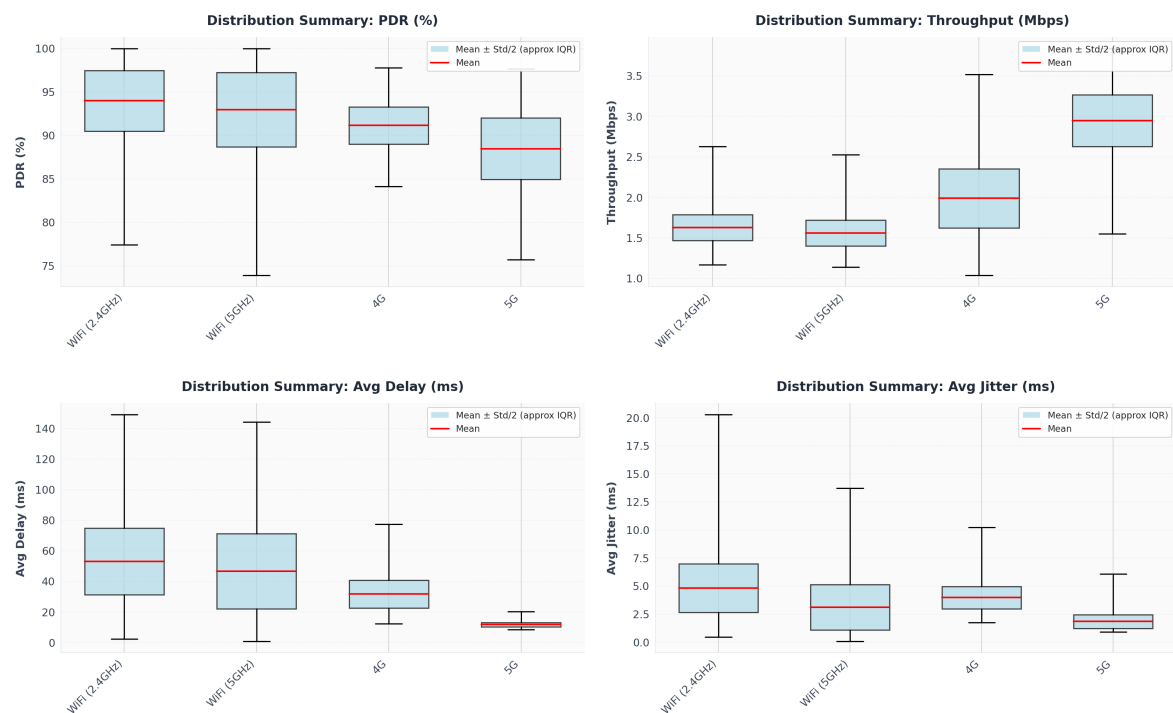
Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	93.22	1.63	46.84
WIFI Mesh (5 GHz)	90.34	1.47	53.11
4G	91.64	1.70	28.69
5G	90.54	3.19	13.27

15 Nodes

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	89.86	1.39	78.31
WIFI Mesh (5 GHz)	88.86	1.43	71.05
4G	87.14	1.46	47.04
5G	91.02	3.05	11.64

3. Statistical Summary

- Tables below summarize distribution of core KPIs across seeds, node counts, and flow scales.



PDR (%)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	93.99	6.99	77.41	100
WIFI Mesh (5 GHz)	92.95	8.56	73.92	100

Technology	Mean	Std Dev	Min	Max
4G	91.15	4.26	84.15	97.77
5G	88.47	7.08	75.72	97.66

Throughput (Mbps)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	1.63	0.32	1.17	2.63
WIFI Mesh (5 GHz)	1.56	0.32	1.14	2.53
4G	1.99	0.73	1.04	3.52
5G	2.95	0.64	1.55	3.84

Avg Delay (ms)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	53.10	43.39	2.31	149.15
WIFI Mesh (5 GHz)	46.68	49.11	0.90	144.11
4G	31.70	18.29	12.32	77.35
5G	11.74	2.96	8.53	20.21

Avg Jitter (ms)

Technology	Mean	Std Dev	Min	Max
WIFI Mesh (2.4 GHz)	4.81	4.33	0.45	20.29
WIFI Mesh (5 GHz)	3.11	4.06	0.09	13.73
4G	3.97	1.96	1.74	10.25
5G	1.85	1.22	0.91	6.08

4. Traffic Load Impact

- WIFI Mesh (2.4 GHz) loses 9.81 PDR points and adds 75.22 ms delay from 10KB to 1MB payload.
- WIFI Mesh (5 GHz) loses 8.92 PDR points and adds 72.08 ms delay from 10KB to 1MB payload.
- 4G gains 2.15 PDR points and adds 24.87 ms delay from 10KB to 1MB payload.
- 5G gains 5.36 PDR points and reduces 0.04 ms delay from 10KB to 1MB payload.



10KB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	99.12	1.45	13.70
WIFI Mesh (5 GHz)	98.54	1.40	3.27
4G	89.71	1.98	16.83
5G	85.24	2.67	11.85

50KB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	93.54	1.68	56.67
WIFI Mesh (5 GHz)	90.70	1.55	61.40
4G	91.87	2.00	36.57
5G	89.56	3.02	11.56

1MB Payload

Technology	Mean PDR (%)	Mean Throughput (Mbps)	Mean Delay (ms)
WIFI Mesh (2.4 GHz)	89.31	1.78	88.92
WIFI Mesh (5 GHz)	89.62	1.72	75.35
4G	91.86	1.99	41.70
5G	90.60	3.15	11.81